

## MACDILL AFB, FL, USA

WMO: 747880

Lat: 27.850N

Lon: 82.517W

Elev: 4

StdP: 101.27

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 12810

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 3.9        | 6.2 | -3.5                          | 2.8 | 8.0  | -1.1 | 3.4 | 10.2 | 9.3                   | 17.6 | 8.4 | 18.0 | 2.8                   | 10   | 0.383 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 8.1                    | 34.1            | 26.1 | 33.5 | 26.0 | 32.8 | 25.8 | 28.2                | 31.6 | 27.7 | 31.2 | 27.3 | 30.7 | 3.4                  | 300  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 27.4                            | 23.2 | 29.6 | 26.9 | 22.5 | 29.3 | 26.3 | 21.8 | 29.0 | 90.0          | 31.7 | 87.8 | 31.3 | 85.7 | 30.8 | 30.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 8.6               | 7.6  | 7.0 | 1.0                        | 35.4 | 2.5                | 0.7 | -0.8                                               | 35.9 | -2.3       | 36.3 | -3.6       | 36.7 | -5.4       | 37.2 |  |  |
|                   |      |     | -1.2                       | 28.6 | 2.9                | 0.8 | -3.2                                               | 29.1 | -4.9       | 29.6 | -6.5       | 30.0 | -8.6       | 30.5 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 23.2   | 15.9  | 17.8  | 19.9  | 22.8  | 25.9  | 27.9  | 28.2  | 28.4  | 27.7  | 24.8  | 20.4  | 18.1 | (6)  |
| (7)  |                                                                                  | DBStd       | 5.33   | 4.63  | 4.12  | 3.56  | 2.69  | 1.95  | 1.31  | 1.20  | 1.19  | 1.39  | 2.83  | 3.60  | 4.47 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 13     | 8     | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 3    | (8)  |
| (9)  |                                                                                  | HDD18.3     | 272    | 102   | 54    | 27    | 4     | 0     | 0     | 0     | 0     | 0     | 2     | 22    | 62   | (9)  |
| (10) |                                                                                  | CDD10.0     | 4822   | 191   | 220   | 307   | 383   | 492   | 538   | 566   | 572   | 531   | 457   | 312   | 254  | (10) |
| (11) |                                                                                  | CDD18.3     | 2039   | 26    | 39    | 76    | 137   | 233   | 288   | 307   | 313   | 281   | 201   | 84    | 54   | (11) |
| (12) |                                                                                  | CDH23.3     | 18795  | 121   | 195   | 437   | 977   | 2153  | 3049  | 3346  | 3411  | 2736  | 1649  | 472   | 249  | (12) |
| (13) | CDH26.7                                                                          | 6982        | 12     | 26    | 77    | 261   | 816   | 1275  | 1377  | 1399  | 1069  | 546   | 87    | 36    | (13) |      |
| (14) | Wind                                                                             | WSAvg       | 3.4    | 3.4   | 3.5   | 3.8   | 3.8   | 3.6   | 3.1   | 2.8   | 2.9   | 3.3   | 3.5   | 3.4   | 3.4  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 1276   | 64    | 68    | 77    | 61    | 64    | 176   | 213   | 219   | 170   | 65    | 39    | 60   | (15) |
| (16) |                                                                                  | PrecMax     | 1769   | 139   | 295   | 232   | 219   | 226   | 357   | 385   | 346   | 346   | 167   | 123   | 396  | (16) |
| (17) |                                                                                  | PrecMin     | 981    | 10    | 11    | 8     | 5     | 6     | 78    | 101   | 118   | 44    | 9     | 6     | 13   | (17) |
| (18) |                                                                                  | PrecStd     | 203    | 39    | 59    | 60    | 43    | 54    | 76    | 66    | 70    | 80    | 35    | 29    | 96   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 28.0  | 29.1  | 30.2  | 31.9  | 34.1  | 34.7  | 35.0  | 34.4  | 32.9  | 30.4  | 29.0  | (19) |      |
| (20) |                                                                                  |             | MCWB   | 21.8  | 21.7  | 21.6  | 22.9  | 23.9  | 25.7  | 26.3  | 26.9  | 26.2  | 25.2  | 24.3  | 23.2 | (20) |
| (21) |                                                                                  | 2%          | DB     | 26.3  | 27.4  | 28.7  | 30.6  | 32.6  | 33.8  | 33.8  | 34.0  | 33.5  | 32.0  | 28.9  | 27.7 | (21) |
| (22) |                                                                                  |             | MCWB   | 20.9  | 21.1  | 21.8  | 22.4  | 23.7  | 25.7  | 26.4  | 26.7  | 26.0  | 24.9  | 23.1  | 22.1 | (22) |
| (23) |                                                                                  | 5%          | DB     | 24.6  | 25.9  | 27.4  | 29.4  | 31.8  | 32.9  | 32.9  | 33.0  | 32.7  | 31.0  | 27.6  | 26.2 | (23) |
| (24) |                                                                                  |             | MCWB   | 19.9  | 20.4  | 21.3  | 22.0  | 23.4  | 25.6  | 26.3  | 26.5  | 25.8  | 24.4  | 21.9  | 21.5 | (24) |
| (25) |                                                                                  | 10%         | DB     | 22.6  | 24.1  | 26.1  | 28.1  | 30.8  | 32.2  | 32.2  | 32.3  | 32.0  | 29.9  | 26.2  | 24.5 | (25) |
| (26) |                                                                                  |             | MCWB   | 19.2  | 19.5  | 20.4  | 21.5  | 23.1  | 25.5  | 26.2  | 26.3  | 25.7  | 23.8  | 21.0  | 20.8 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 23.5  | 23.6  | 25.0  | 26.0  | 26.5  | 28.4  | 28.5  | 28.9  | 28.5  | 27.6  | 26.7  | 25.2 | (27) |
| (28) |                                                                                  |             | MCD B  | 26.1  | 26.6  | 27.3  | 29.1  | 30.4  | 31.8  | 31.8  | 32.2  | 31.3  | 30.1  | 28.7  | 27.4 | (28) |
| (29) |                                                                                  | 2%          | WB     | 22.0  | 22.6  | 23.8  | 24.8  | 25.7  | 27.6  | 27.9  | 28.2  | 27.8  | 26.6  | 24.5  | 23.5 | (29) |
| (30) |                                                                                  |             | MCD B  | 24.8  | 25.6  | 26.2  | 27.7  | 29.7  | 31.1  | 31.5  | 31.9  | 30.9  | 29.5  | 26.8  | 25.7 | (30) |
| (31) |                                                                                  | 5%          | WB     | 21.0  | 21.6  | 22.8  | 23.9  | 25.2  | 27.0  | 27.4  | 27.7  | 27.1  | 25.8  | 23.5  | 22.6 | (31) |
| (32) |                                                                                  |             | MCD B  | 23.3  | 24.4  | 25.3  | 26.9  | 29.1  | 30.7  | 31.1  | 31.3  | 30.4  | 28.8  | 26.0  | 24.7 | (32) |
| (33) |                                                                                  | 10%         | WB     | 20.0  | 20.7  | 21.8  | 23.1  | 24.6  | 26.4  | 26.9  | 27.2  | 26.5  | 25.2  | 22.5  | 21.7 | (33) |
| (34) |                                                                                  |             | MCD B  | 22.0  | 23.0  | 24.4  | 26.3  | 28.4  | 30.4  | 30.4  | 30.6  | 29.8  | 28.1  | 24.9  | 23.7 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 10.7  | 10.3  | 10.3  | 10.1  | 9.8   | 8.6   | 8.1   | 8.1   | 8.4   | 9.5   | 10.3  | 10.1 | (35) |
| (36) |                                                                                  |             | MCDBR  | 11.3  | 11.1  | 11.0  | 11.2  | 10.7  | 9.5   | 9.0   | 9.0   | 9.4   | 9.8   | 10.3  | 10.5 | (36) |
| (37) |                                                                                  |             | MCWBR  | 6.5   | 5.8   | 5.3   | 4.8   | 4.2   | 3.6   | 3.7   | 3.6   | 3.7   | 4.2   | 5.3   | 5.6  | (37) |
| (38) |                                                                                  | 5% WB       | MCD BR | 9.8   | 9.3   | 8.9   | 9.3   | 8.8   | 8.6   | 8.5   | 8.5   | 8.7   | 8.5   | 8.6   | 9.2  | (38) |
| (39) |                                                                                  |             | MCWBR  | 6.7   | 5.8   | 5.6   | 5.4   | 4.0   | 4.0   | 4.0   | 4.0   | 4.1   | 4.4   | 5.3   | 5.8  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.342  | 0.355 | 0.368 | 0.392 | 0.424 | 0.449 | 0.473 | 0.453 | 0.422 | 0.390 | 0.354 | 0.343 | (40) |      |
| (41) |                                                                                  | taud        | 2.523  | 2.495 | 2.445 | 2.341 | 2.310 | 2.321 | 2.258 | 2.327 | 2.382 | 2.456 | 2.501 | 2.548 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 903    | 918   | 923   | 905   | 871   | 844   | 823   | 839   | 858   | 869   | 881   | 887   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 91     | 101   | 112   | 127   | 132   | 130   | 138   | 128   | 118   | 103   | 92    | 85    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 3.55   | 4.29  | 5.42  | 6.50  | 6.88  | 6.21  | 5.88  | 5.66  | 5.15  | 4.76  | 3.88  | 3.29  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.25   | 0.27  | 0.44  | 0.31  | 0.38  | 0.50  | 0.36  | 0.35  | 0.33  | 0.29  | 0.20  | 0.27  | (45) |      |

## Historical Trends

|      |                         | Heating |        | Cooling |       |       | Degree-Days |         |         |         |      |      |
|------|-------------------------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|------|------|
|      |                         | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |      |      |
|      |                         | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     | (m)  |      |
| (46) | Station Only            | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     | N/A  | (46) |
| (47) | Regional (35 neighbors) | +0.42   | N/A    | N/A     | N/A   | N/A   | N/A         | -3      | -45     | +173    | +121 | (47) |

Nomenclature: See separate page